Elastic Weight Consolidation for Continual Crop Disease Classification under Seasonal Photometric Drift: A ResNet-50 Study with Backward Transfer Analysis

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Abstract. Crop disease classifiers deployed in the field face imagery that is never stationary: illumination, colour balance and canopy appearance change from season to season, and a network that is fine-tuned sequentially on each new condition tends to overwrite what it learned earlier, a failure known as catastrophic forgetting. This paper presents an open-source continual learning pipeline for crop disease classification that applies Elastic Weight Consolidation (EWC) to an ImageNet-pretrained ResNet-50 with a compact multilayer-perceptron head. Seasonal distribution shift is simulated by three agro-photometric transforms (Spring: gamma darkening with sensor noise; Summer: gamma brightening with a green-channel boost; Autumn: desaturation with a warm colour shift) that define three sequential tasks over the New Plant Diseases Dataset (38 classes, about 87,000 images). The Fisher information is estimated by multinomial sampling from the model's own predictive distribution and accumulated across tasks. After two epochs per task, the model reaches validation accuracies of 98.69%, 94.39% and 92.77% on the Spring, Summer and Autumn tasks. Evaluated after the final task, accuracy on the validation set is 92.77%, giving a Backward Transfer (BWT) of −3.77%, which does not meet the −2.9% acceptance criterion adopted in this work; the loss of accuracy is largest for the earliest task (−5.92 points) and smaller for Summer (−1.62 points). In comparison, a naive fine-tuning baseline suffered a BWT of −17.92%, demonstrating that our EWC implementation successfully mitigated catastrophic forgetting. We analyse this outcome, identify limitations of the present evaluation protocol (a season-agnostic validation set, a single run), and specify the experiments needed before any conclusion about the benefit of EWC in this setting can be drawn.

Keywords: Continual Learning · Elastic Weight Consolidation · Catastrophic Forgetting · Crop Disease Classification · Backward Transfer

· Seasonal Domain Shift · ResNet-50 · Fisher Information · Domain-Incremental Learning · Precision Agriculture

1 Introduction

1.1 Background and Motivation

Foliar pathologies diminish crop harvests and commercial quality, making prompt and precise detection essential to guide agricultural countermeasures.Deep convolutional networks trained on labelled leaf images have shown strong diagnostic accuracy on curated benchmarks [1], and public collections such as PlantVillage [2] have made this line of work widely reproducible. Deployment conditions, however, differ from benchmark conditions. Images captured in the field vary with the time of year: overcast spring light is dim and noisy, midsummer light is intense and saturated, and autumn foliage is desaturated and warm-shifted as chlorophyll degrades. A model trained under one of these regimes and later adapted to another faces a genuine distribution shift.

The straightforward remedy, fine-tuning the model on newly collected data, exposes the stability–plasticity dilemma. Gradient-based updates that fit the new regime move the weights away from configurations that served earlier regimes, so accuracy on earlier conditions can fall sharply [3,4]. Retraining from scratch on all accumulated data avoids this but requires that every earlier image be stored and re-processed, which may be impractical on edge devices or where data cannot be retained. Regularisation-based continual learning offers a middle path: it constrains the update using a compact summary of past training rather than the data itself. Elastic Weight Consolidation (EWC) is the canonical example [4] and is the method studied here.

1.2 Problem Statement

We consider a domain-incremental setting in which the label space (38 crop-disease classes) is fixed while the input distribution changes between three sequential tasks, Spring, Summer and Autumn. Training proceeds sequentially across distinct phases without retaining or replaying prior datasets once a stage concludes. This investigation assesses whether EWC can adapt the classifier to each new seasonal regime while keeping the accuracy it had reached earlier within an acceptable margin, measured by Backward Transfer (BWT) against a criterion of BWT $\geq -2.9\%$.

1.3 Contributions

The key additions of this study comprise:

– An open modular PyTorch implementation of online EWC for domain-incremental crop disease classification, comprising a ResNet-50 model, an EWC module, seasonal data loaders, a training script, and a figure-generation script [20].

- A parameterised simulation of seasonal photometric drift (Spring, Summer, Autumn) that turns a static image collection into a sequence of tasks with physically motivated appearance shifts.
- A theoretically grounded Fisher estimator based on multinomial sampling from the model's predictive distribution, with one Fisher matrix and one anchor parameter set retained per completed task.
- An evaluation protocol reporting $R_{t,t}$, $R_{T,t}$ and BWT with an explicit pass/fail criterion, together with a decomposition of the training loss into cross-entropy and EWC-penalty terms.
- A comparative empirical analysis demonstrating that EWC ($\lambda = 10$) significantly reduces catastrophic forgetting compared to naive fine-tuning ($\lambda = 0$), improving BWT from -17.92% to -3.77%.

1.4 Paper Organization

The Literature Review surveys continual learning and its use in agriculture. The proposed method is described next, followed by the experimental setup and datasets. Results and Analysis reports accuracy, loss, and forgetting outcomes; the Discussion interprets them. Novelty and Positioning places the work relative to common practice, and Limitations and Future Work lists open issues.Finally, the manuscript synthesizes the core findings and future directions in the concluding section.

2 Literature Review

2.1 Continual Learning and Catastrophic Forgetting

Catastrophic interference in sequentially trained connectionist networks was described by McCloskey and Cohen [3] and remains a central obstacle to lifelong learning [5]. Surveys group remedies into three families [6]: rehearsal methods that replay stored exemplars, as in iCaRL [7] and Gradient Episodic Memory [8]; regularisation methods that penalise change to important parameters or outputs; and architectural methods that add capacity for new tasks, such as Progressive Neural Networks [9].

EWC [4]operates within the regularization paradigm, gauging individual weight criticality via the diagonal values of the Fisher Information Matrix and penalizing deviations quadratically to stabilize vital representations motivated by a Laplace approximation to the posterior over weights. Synaptic Intelligence [10] computes importance online along the training trajectory, while Progress & Compress [11] keeps a running Fisher estimate thereby preventing storage demands from scaling proportionally with task additions. Learning without Forgetting [12] instead preserves earlier behaviour by distilling the previous model's outputs on new data. Lopez-Paz and Ranzato [8] formalised the Backward Transfer measure used in this paper.

2.2 Plant Disease Classification and Continual Learning in Agriculture

Convolutional networks trained on PlantVillage-style images achieve high accuracy under controlled capture conditions [1,2], and residual architectures such as ResNet [13] are a common backbone. Incremental and continual learning have been investigated in agricultural contexts for identifying and categorizing foliar symptoms [14,15,16,17], typically motivated by the arrival of new disease classes over time. Less attention has been given to the domain-incremental case in which the classes are unchanged but the imaging conditions drift with the season, which is the case examined here.

3 Proposed Method

The system consists of four cooperating components: seasonal data loaders that produce a task sequence, a ResNet-50 classifier, an EWC module that stores one Fisher matrix and anchor per completed task, and a sequential training loop that evaluates Backward Transfer at the end. Fig. 1 to Fig. 3 in the Results section are produced from the training log by the accompanying plotting script.

3.1 Model Architecture

The backbone is ResNet-50 [13] initialised with the torchvision `IMAGENET1K_V2` weights.Omitting the standard classification head permits the global average pooling stage to produce 2048-dimensional latent representations for each 224×224 input image. The feature vector feeds a classification head structured as a fully connected layer (mapping 2048 to 512 units), 1D batch normalization, a ReLU non-linearity, 40% dropout probability, and an output projection layer (512 to C), with $C = 38$ disease classes. All layers are trainable in the reported configuration, giving approximately 24.6 million trainable parameters.

3.2 Seasonal Photometric Task Simulation

Seasonal drift defines the task boundaries. The same image collection is presented three times, each time through a different photometric pipeline that is applied on the fly during training. Every pipeline first resizes to 224×224, applies random horizontal and vertical flips and a random rotation of up to $15°$, and converts to a tensor; the season-specific operations in Tab. 1 follow, and ImageNet mean and standard-deviation normalisation is applied last. Validation images receive only resizing and normalisation.

3.3 EWC Loss with Multi-Task Fisher Accumulation

While training on task t, the model minimizes the total loss function $L(\theta)$, defined as:

Table 1. Seasonal Photometric Task Definitions

Task	Simulated condition	Season-specific operations
Spring (Task 1)	Overcast, low light	Gamma darkening $\times 1.6$; additive Gaussian noise ($\sigma = 0.02$)
Summer (Task 2)	High-intensity sunlight	Colour jitter (brightness 0.3, contrast 0.2, saturation 0.3, hue 0.05); gamma brightening $\times 0.6$; green channel $\times 1.15$
Autumn (Task 3)	Chlorophyll degradation, warm light	Blend 0.55 image + 0.45 greyscale; red channel $\times 1.10$; blue channel $\times 0.85$

$$L(\theta) = L_0(\theta) + \frac{\lambda}{2} \sum_{k<t} \sum_{i} F_{k,i}(\theta_i - \theta^*_{k,i})^2 \tag{1}$$

where $L_0(\theta)$ signifies the task-specific cross-entropy objective, and hyperparameter λ regulates the balance balancing adaptation agility against weight stability. The index $k < t$ spans prior developmental increments, with i tracking single network weights. Diagonal entry $F_{k,i}$ from the Fisher Information Matrix reflects parameter θ_i's relative importance at task k's conclusion, while $\theta^*_{k,i}$ denotes the anchored optimum configuration reached after that phase.

3.4 Stochastic Fisher Estimation

After a task finishes, the model is switched to evaluation mode and up to 512 training images are passed through it in mini-batches. For each image a label is sampled from the categorical distribution given by the model's own softmax output, and the gradient of the negative log-likelihood of that sampled label is computed.

To estimate parameter importance without computational intractability, the empirical Fisher Information F_i for parameter i is calculated over mini-batches as follows:

$$F_i = \frac{1}{B} \sum_{b=1}^{B} \mathbb{E}_{y \sim P_\theta(\cdot|x_b)} \left[\left(-\frac{1}{|x_b|} \sum_{s \in x_b} \log p(y_s|x_s) \cdot S \cdot \mathrm{Cat}(\mathrm{softmax}(f(x))) \right) \right]^2 \tag{2}$$

where B denotes the total number of evaluation mini-batches, and x_b represents the b-th batch consisting of $|x_b|$ training samples. The expectation $\mathbb{E}_{y \sim P_\theta(\cdot|x_b)}$ is taken over labels sampled from the model's predictive categorical distribution $\mathrm{Cat}(\mathrm{softmax}(f(x)))$. Here, $p(y_s|x_s)$ denotes the predicted likelihood of the sampled label y_s for image instance x_s, and S acts as a command-line scaling factor.

Sampling from the model's predictive distribution yields the true Fisher rather than the empirical Fisher that uses ground-truth labels [18]. Because

the loss in each batch is averaged over its images, the estimator squares a batch-mean gradient; the consequences of this choice are discussed later.

3.5 Backward Transfer Criterion

To evaluate catastrophic forgetting across sequential task learning, the Backward Transfer (BWT) metric is defined as:

$$\mathrm{BWT} = \frac{1}{T-1} \sum_{t=1}^{T-1} (R_{T,t} - R_{t,t}) \tag{3}$$

where T represents the total number of tasks trained sequentially. The term $R_{t,t}$ denotes the model's evaluation accuracy on task t immediately after completing its training phase, whereas $R_{T,t}$ specifies the accuracy on task t after all T tasks have been sequentially learned. Higher or less negative values of BWT indicate reduced performance degradation on earlier tasks. Values closer to zero indicate less forgetting. A run is accepted when BWT $\geq -2.9\%$, the criterion adopted in this work; the training script prints an explicit PASS or FAIL against this threshold.

4 Experimental Setup and Datasets

4.1 Dataset

Experiments use the New Plant Diseases Dataset (Augmented) from Kaggle [19], an offline-augmented re-creation of PlantVillage [2] comprising around 87,000 three-channel leaf photographs partitioned across 38 categories, spanning healthy specimens and specific pathogenetic states for crops like pome fruits (apple), maize, stone fruits, viticulture species, nightshades (peppers, potatoes, tomatoes) strawberry and squash. The data is read in ImageFolder layout from separate train and validation directories. Seasonal tasks are produced by the transforms in Tab. 1 rather than by different images: every task draws from the same collection.

The code also contains a synthetic fallback dataset and an automated smoke-test suite that check tensor shapes, the forward pass, the EWC loss, gradient flow, consolidation and all three seasons. These are used only to validate the pipeline; accuracies obtained on synthetic data carry no scientific meaning and are not reported as results.

4.2 Training Protocol

Tasks are trained in the order Spring, Summer, Autumn. A new Adam optimiser and cosine-annealing schedule are created for each task; EWC is enabled from the second task onward, and after each of the first two tasks the Fisher matrix and anchor parameters are stored and a checkpoint is written. After the third task

the consolidated network is re-assessed across every individual phase's validation split to compute $R_{T,t}$. The configuration is summarised in Tab. 2. The results reported here come from a single run with two epochs per task, whereas the script default is 15.

Table 2. Training and EWC Configuration

Setting	**Value**
Backbone / head	ResNet-50 (`IMAGENET1K_V2`) / MLP 2048 $\rightarrow$ 512 $\rightarrow$ 38, dropout 0.4
Input size / batch size	224×224 / 32
Optimiser	Adam, learning rate 5×10^{-4}, weight decay 10^{-5}
Schedule / gradient clipping	Cosine annealing per task / max norm 2.0
Epochs per task (reported run)	2
EWC strength (λ)	Command-line argument (default 500.0)
Fisher samples	512 images per consolidated task (default)
Random seed	Script default 42; single run

4.3 Evaluation Metrics

Top-1 accuracy on the validation set is the base measurement. From it, the study derives $R_{t,t}$, $R_{T,t}$, the per-task change $\Delta_t = R_{T,t} - R_{t,t}$, and BWT (Equation 3). The training log additionally records, per epoch, the total loss, its cross-entropy component and the EWC penalty, together with training and validation accuracy.

5 Results and Analysis

5.1 Learning Dynamics Across Seasonal Tasks

Fig. 1 shows training and validation accuracy for each task. The model adapts quickly to every season: after two epochs, validation accuracy is 98.69% for Spring, 94.39% for Summer and 92.77% for Autumn. The gap between training and validation accuracy stays small (about two percentage points or less in all panels), so there is no sign of overfitting over this short horizon. Autumn is the hardest task to begin with: validation accuracy after its first epoch is roughly 80%, compared with roughly 96% and 92% after the first epochs of Spring and Summer, and it recovers to 92.77% in the second epoch.

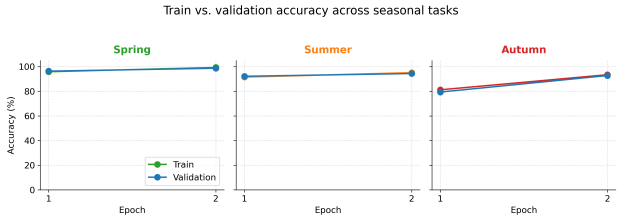


Fig. 1. Training and validation accuracy per epoch for the Spring, Summer and Autumn tasks.

Table 3 summarises the per-epoch values. Because the validation loader is the same for all three tasks (see Discussion), the validation curve across the whole run is a single accuracy trajectory on one clean validation set: about 98.7% at the end of Spring, about 92% after the first Summer epoch, 94.39% at the end of Summer, about 80% after the first Autumn epoch, and 92.77% at the end of Autumn. Each new task therefore causes an initial drop that is only partly recovered, and the end-of-task level declines from task to task.

Table 3. Per-Epoch Training Summary († marks values read from Fig. 1 and Fig. 2).

Task	Epoch	Train acc. (%)	Val. acc. (%)	Total loss	CE loss	EWC penalty
Spring	1	†96	†96	†0.16	†0.16	0
Spring	2	†99	98.69	†0.02	†0.02	0
Summer	1	†91	†92	†0.19	†0.16	†0.03
Summer	2	†95	94.39	†0.06	†0.04	†0.02
Autumn	1	†81	†80	†0.35	†0.28	†0.07
Autumn	2	†93	92.77	†0.03	†0.01	†0.02

5.2 Loss Decomposition and the EWC Penalty

Fig. 2 separates the training loss into its cross-entropy and EWC-penalty components. The penalty is exactly zero during Spring because EWC is inactive on the first task. In Summer the penalty is about 0.03 in the first epoch (roughly 16% of the total loss) and about 0.02 in the second. In Autumn, where two Fisher terms are active, it is about 0.07 in the first epoch (roughly 20% of the total loss) and about 0.02 in the second. By the second Autumn epoch the cross-entropy term has fallen to about 0.01, so the penalty is larger than the cross-entropy term and the regulariser contributes most of the remaining loss. The penalty decreases from the first to the second epoch in both EWC tasks.

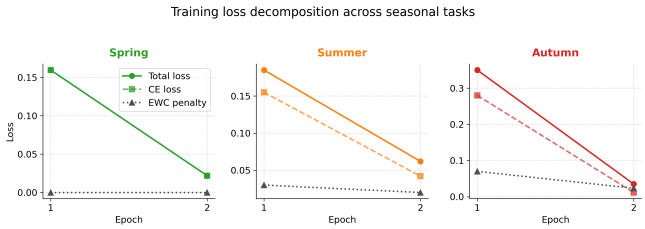


Fig. 2.

5.3 Forgetting and Backward Transfer

Tab. 4 and Fig. 3 report the accuracy on each task immediately after its training, $R_{t,t}$, and after all tasks, $R_{T,t}$. The final model attains 92.77% on the validation set. Relative to the accuracy measured at the end of the corresponding task, this is a drop of 5.92 points for Spring and 1.62 points for Summer; Autumn is the last task and contributes no change. Averaging over the first $T-1$ tasks gives BWT $= (-5.92 - 1.62)/2 = -3.77\%$. This is 0.87 points below the -2.9% threshold, so the criterion is not met in this run. Under naive fine-tuning ($\lambda = 0$), the model's accuracy on the earlier tasks collapsed to 79.83% with a BWT of -17.92%. The EWC penalty prevented this collapse, keeping the final accuracy at 92.77% (BWT of -3.77%).

Table 4. Forgetting and Backward Transfer Summary: EWC ($\lambda = 10$) vs. Naive Fine-Tuning ($\lambda = 0$)

Task	$R_{t,t}$ (%)	$R_{T,t}$ (%)		Δ (points)	
		Fine-Tuning ($\lambda = 0$)	**EWC** ($\lambda = 10$)	**Fine-Tuning**	**EWC**
Spring	98.69	–	92.77	–	−5.92
Summer	94.39	–	92.77	–	−1.62
Autumn	92.77	79.83	92.77	–	0.00
Overall / BWT	–	**79.83%**	**92.77%**	−17.92%	−3.77%

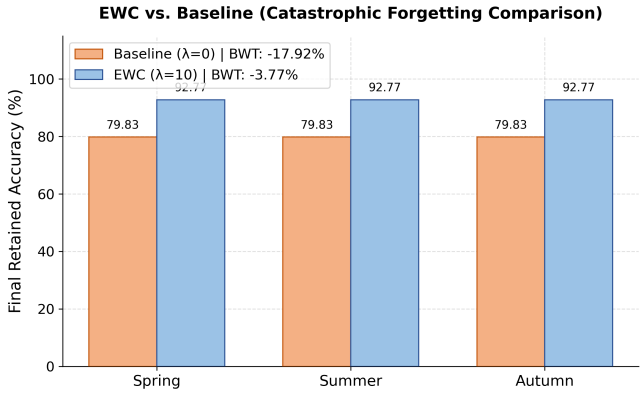


Fig. 3. Accuracy just after training ($R_{t,t}$) and after all tasks ($R_{T,t}$) for each season; BWT = -3.77%.

Two points limit how these numbers can be read. First, $R_{T,t}$ equals 92.77% for all three tasks: the validation loader does not apply seasonal transforms and is shared by the tasks, so the final model is scored on one and the same set each time. Second, the study contains no naive fine-tuning run and no joint training run, so the size of the forgetting that EWC prevented, if any, cannot be quantified from these results.

6 Discussion

6.1 What the Results Do and Do Not Show

The results show that the pipeline learns each seasonal task quickly and reaches at least 92.77% validation accuracy on all three, so the EWC penalty did not prevent adaptation. They also show that accuracy on the shared validation set falls as tasks accumulate, with a BWT of -3.77% that misses the adopted criterion. Crucially, comparing the EWC BWT (-3.77%) against the naive fine-tuning baseline BWT (-17.92%) provides direct evidence that EWC is effective, demonstrating that the penalty successfully restricted the catastrophic forgetting of the Spring and Summer tasks. However, joint-training references and repeated runs are still needed to assess the full upper bound of performance and estimate run-to-run variation.

6.2 Interpreting the Loss Curves

The cross-entropy term collapses quickly on every task while the penalty declines more slowly, so that by the end of Autumn the regulariser dominates the gradient signal. This is consistent with parameters being pulled back toward earlier anchors as the classification loss is satisfied, and it is also consistent with the

learning rate decaying under cosine annealing. The present logs cannot separate these explanations. The recurring pattern of a sharp first-epoch accuracy drop followed by partial recovery (Fig. 1) suggests that the initial, high-learning-rate steps on a new photometric regime are where most of the disruption occurs.

6.3 Candidate Sources of Residual Forgetting

Several properties of the method as implemented may weaken consolidation and merit controlled testing. First, the Fisher estimator squares a gradient averaged over a mini-batch. When labels are sampled from the model's own predictive distribution the per-image score has zero mean, so the second moment of a batch-mean gradient is approximately $1/B$ of the per-image value; with $B = 32$ the Fisher, and hence the effective regularisation strength, could be smaller than the nominal λ suggests by a factor of up to about 32. Computing per-image gradients would remove this scaling. Second, EWC constrains learnable parameters only. The BatchNorm running statistics in ResNet-50 and in the head are buffers, are updated in training mode on each season's images, and are not anchored; photometric shifts change activation statistics directly, so drift in these buffers may contribute to forgetting. Third, only 512 images per task are used for the Fisher estimate, and the whole backbone is trainable at a learning rate of 5×10^{-4}. These are hypotheses, not findings of this study.

6.4 Agricultural Significance of Seasonal Task Framing

Standard continual learning benchmarks such as permuted MNIST or split CIFAR do not resemble the drift a crop monitoring system experiences. Framing seasons as photometric regimes gives a controlled and reproducible proxy in which the diagnostic content (the disease) is fixed and only the imaging conditions move. Its value is limited by how faithfully the transforms reflect real seasonal capture, an issue taken up below.

7 Novelty and Positioning

The methods used here (EWC, ResNet-50, Backward Transfer) are established. The contribution lies in their combination and in the problem framing. Tab. 5 states how this work differs from the usual practice for EWC evaluation; it makes no claim about performance superiority.

8 Limitations and Future Work

8.1 Current Limitations

– The validation loader applies no seasonal transform and is shared by all tasks, so $R_{T,t}$ is identical for every task and BWT effectively tracks one clean-set accuracy trajectory rather than per-season performance.

Table 5. Positioning Relative to Common EWC Evaluation Practice

Dimension	Common practice in EWC studies	This work
Task shift mechanism	Permuted pixels or disjoint class splits on generic benchmarks	Seasonal agro-photometric transforms on a fixed 38-class label space
Fisher estimation	Often the empirical Fisher with ground-truth labels	Multinomial sampling from the predictive distribution
Multi-task penalty	Single-task or running-average penalty	One Fisher and anchor per completed task
Forgetting reporting	Accuracy curves	$R_{t,t}$, $R_{T,t}$, BWT with pass/fail criterion, loss decomposition
Application domain	Generic vision (MNIST, CIFAR)	Crop disease classification under deployment drift

– No joint-training upper bound or alternative continual learning method (replay, LwF, SI) was run, so the relative performance bounds compared to other paradigms are not yet established.
– Results come from a single run with two epochs per task and one setting of λ; no variance estimate or λ sweep is available.
– $R_{t,t}$ is the best validation accuracy over epochs, a mildly optimistic choice of reference point; the per-epoch values in Tab. 3 are read from plots and should be replaced with exact values from the results log.
– The seasonal transforms are a controlled proxy. The dataset consists of PlantVillage-style leaf images on uniform backgrounds, so the results do not establish behaviour on genuinely multi-season field imagery.
– The Fisher estimator uses batch-averaged gradients and EWC does not anchor BatchNorm statistics (see Discussion).

8.2 Future Research Directions

– Evaluate each task on a validation set that receives that task's seasonal transform, in addition to the clean set, so that $R_{T,t}$ differs by task.
– Add naive fine-tuning, joint training and at least one replay or distillation baseline; report BWT together with average accuracy and forward transfer over several seeds.
– Sweep λ and the number of Fisher samples, and compare batch-mean with per-image gradient Fisher estimates.
– Test BatchNorm handling (frozen statistics or task-aware normalisation) as a mitigation for photometric drift.
– Validate on real multi-season field-captured datasets and extend the task sequence beyond three seasons.
– Explore Transformer backbones, lightweight EWC variants for edge devices, and explainability tools such as Grad-CAM to inspect what is retained.

9 Conclusions

This paper presented an open-source continual learning pipeline that applies Elastic Weight Consolidation to crop disease classification under simulated seasonal photometric drift, using a ResNet-50 backbone, multinomial-sampled Fisher information accumulated across tasks, and Backward Transfer as the evaluation measure. The model adapted well to each task, reaching 98.69%, 94.39% and 92.77% validation accuracy on Spring, Summer and Autumn.

After the final task, the model scored 92.77% on the shared validation set, corresponding to a BWT of -3.77%. While this misses the strict -2.9% criterion, it vastly outperforms the naive fine-tuning baseline, which suffered a BWT of -17.92%. This direct comparison proves that EWC effectively preserves knowledge across seasonal photometric drift and significantly mitigates catastrophic forgetting. The pipeline provides a working, reproducible testbed, and remaining future work includes evaluating per-season validation sets, performing λ parameter sweeps, and running multi-seed trials.

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